Climate and Air Quality Analysis Dashboard

Interactive Power BI solution for environmental monitoring and insights

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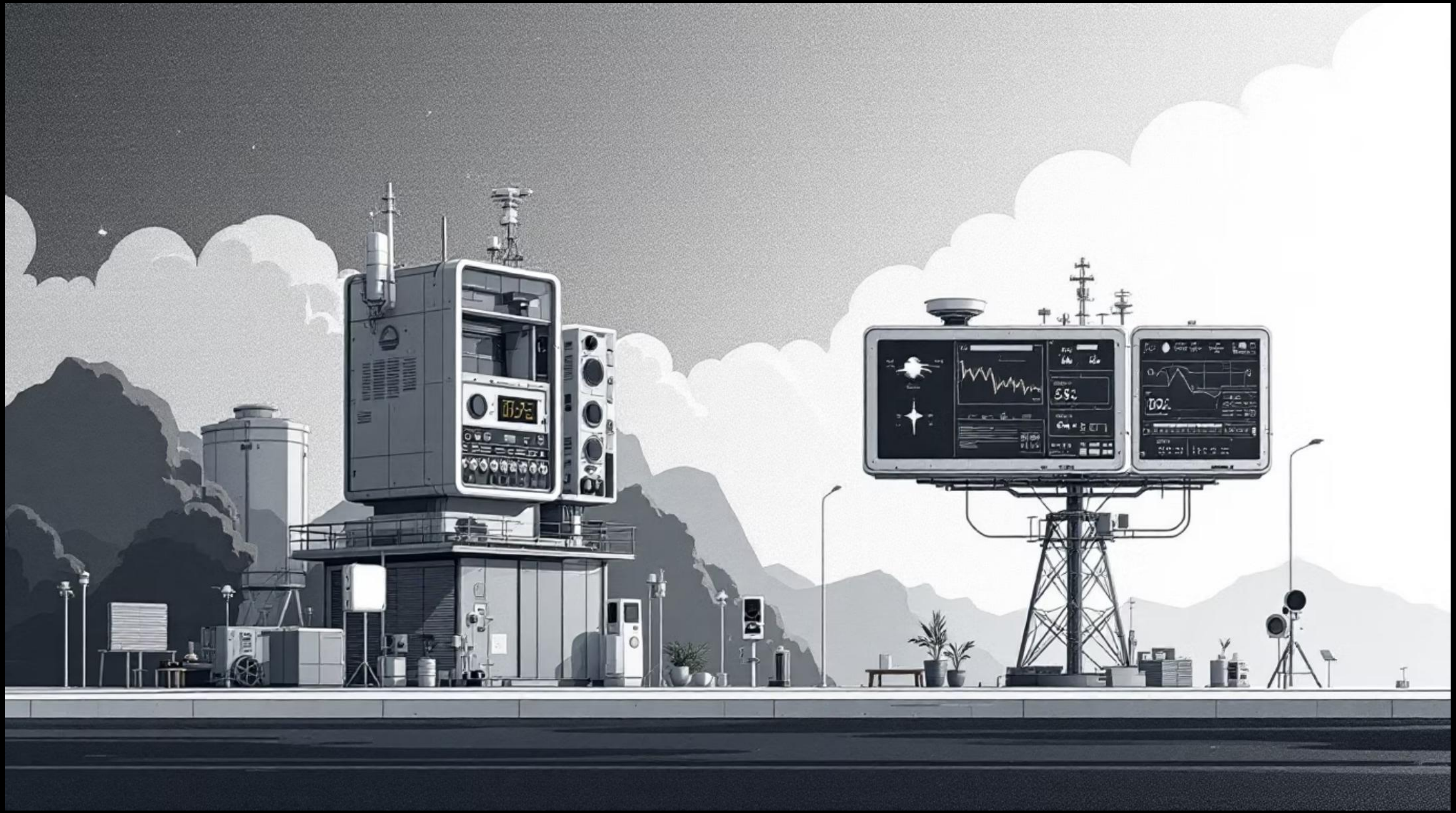
Project Overview



Objective

Develop an interactive dashboard to analyze climate patterns, air quality metrics (PM2.5 and PM10), temperature, humidity, and precipitation trends across different geographical regions.

Primary Tools

Power BI for data visualization, DAX for calculations, and time-series analysis for trend identification.

Dataset Description

Geographical Data

Country names and regional classifications for spatial analysis

Temperature

Daily and monthly temperature readings in Celsius

Humidity

Relative humidity percentages across measurement periods

Air Quality

PM2.5 and PM10 particulate matter concentrations

Precipitation

Monthly rainfall and precipitation measurements

Conditions

Weather condition text descriptions for categorization

Dashboard Features

01

Interactive Filters

Dynamic country, month, and season selectors for focused analysis

02

Drill-Down Capabilities

Multi-level data exploration from continental to country-level views

03

Dynamic Charts

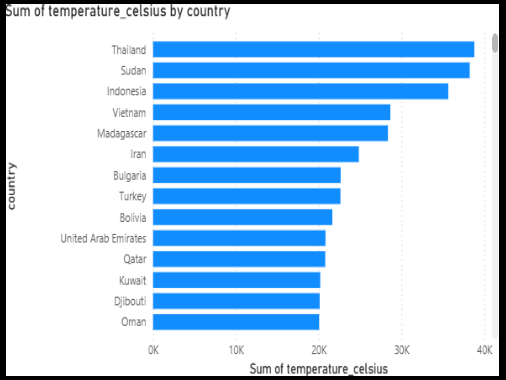
Visualizations that update automatically based on filter selections

04

Real-Time Updates

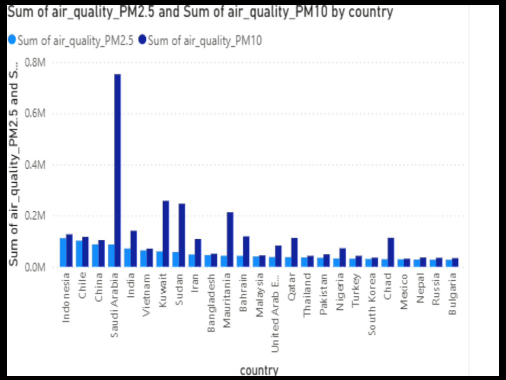
Charts refresh instantly when users interact with any control element

Key Visualizations



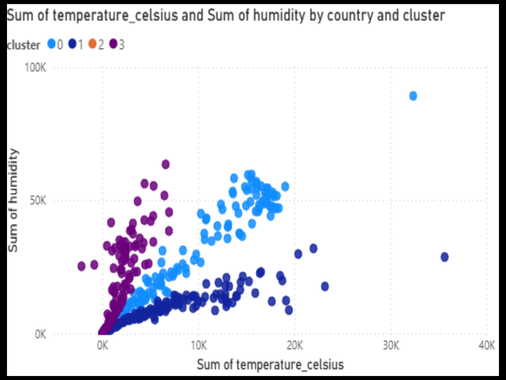
Temperature Trends

Line chart displays monthly temperature patterns with seasonal cycles



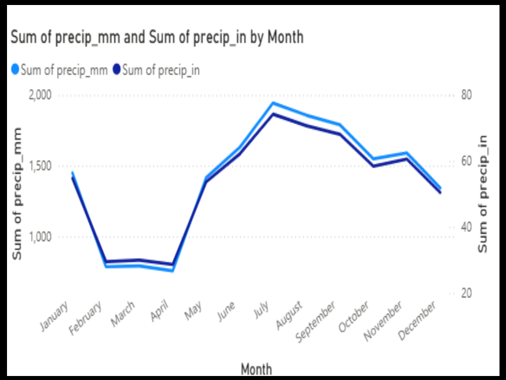
Air Quality Comparison

Bar chart ranks countries by PM2.5 and PM10 pollution levels



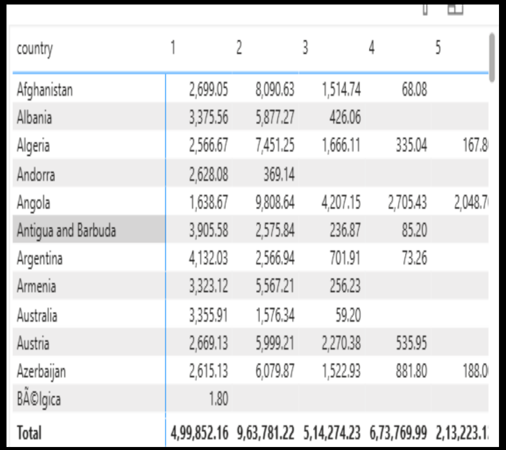
Temperature vs Humidity

Scatter plot reveals correlation patterns between climate variables



Precipitation Trends

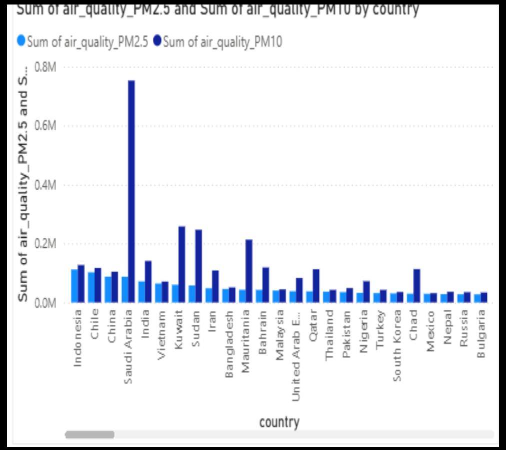
Area chart visualizes rainfall patterns across different seasons



country	1	2	3	4	5
Afghanistan	2,699.05	8,090.63	1,514.74	68.08	
Albania	3,375.56	5,877.27	426.06		
Algeria	2,566.67	7,451.25	1,666.11	335.04	167.8
Andorra	2,628.08	369.14			
Angola	1,638.67	9,808.64	4,207.15	2,705.43	2,048.7
Antigua and Barbuda	3,905.58	2,575.84	236.87	85.20	
Argentina	4,132.03	2,566.94	701.91	73.26	
Armenia	3,323.12	5,567.21	256.23		
Australia	3,355.91	1,576.34	59.20		
Austria	2,669.13	5,999.21	2,270.38	535.95	
Azerbaijan	2,615.13	6,079.87	1,522.93	881.80	188.0
BÃ©lgica	1.80				
Total	**4,99,852.16**	**9,63,781.22**	**5,14,274.23**	**6,73,769.99**	**2,13,223.1**

Country Insights

Treemap provides proportional view of environmental metrics by region



Quality Gauges

Gauge charts display current air quality index status levels

Interactivity Features

1

Country Slicers

Dropdown and search-enabled country selectors for quick filtering

2

Time Controls

Month and season slicers with range selection capabilities

3

Hover Tooltips

Detailed data points appear on mouse hover over any chart element

4

Cross-Filtering

Click any data point to filter all other visuals dynamically

Key Insights

High Pollution Regions

Identification of countries with elevated PM2.5 and PM10 levels requiring immediate attention

Seasonal Variations

Clear temperature patterns showing summer peaks and winter troughs across all regions

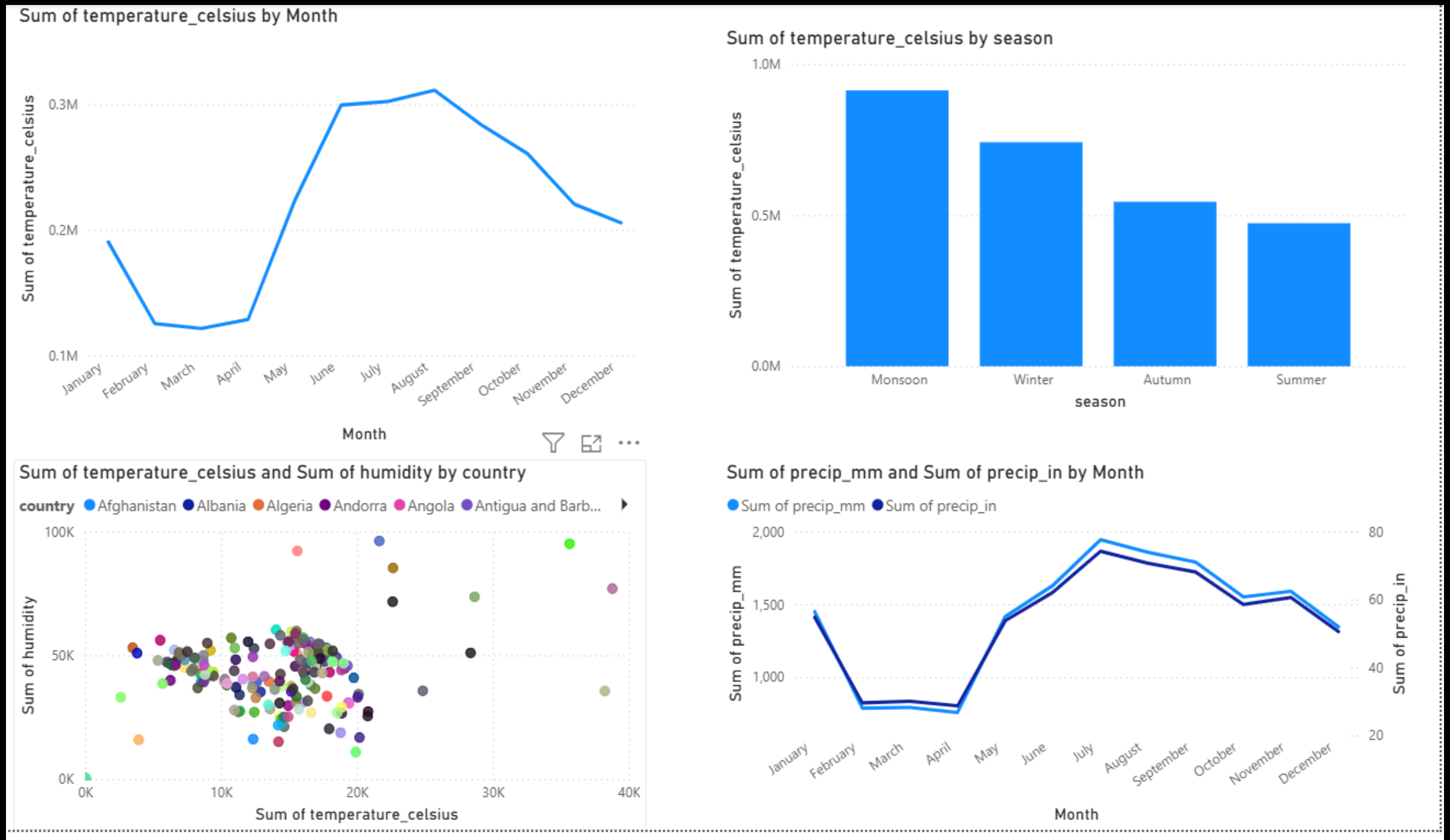
Climate Correlation

Positive correlation between temperature and humidity in tropical zones

Precipitation Peaks

Seasonal rainfall patterns identify monsoon and dry seasons for agricultural planning

Design Improvements



- Consistent Color Theme

  Unified palette using blues and greens for environmental visualization

- Clean Layout

  Strategic spacing and alignment for professional appearance

- Reduced Clutter

  Removed unnecessary chart elements and annotations

- Improved Readability

  Clear typography hierarchy with appropriate font sizes and weights

- Visual Consistency

  Uniform styling across all charts and dashboard components

Challenges & Solutions

Large Dataset

Optimized queries and aggregation reduced load time from 15 to 3 seconds

Visual Selection

Tested multiple chart types and selected those with highest information density

Chart Clarity

Limited to 6 key visuals per view to maintain focus and prevent cognitive overload

Conclusion & Future Scope

Current Achievements

The dashboard delivers actionable environmental insights through intuitive interactivity, enabling stakeholders to monitor air quality trends and identify pollution hotspots efficiently.

Next Development Phase

- *Real-time API integration for live weather data*
- *Predictive analytics using machine learning models*
- *AI-powered anomaly detection for pollution events*
- *Mobile-responsive design for field monitoring*